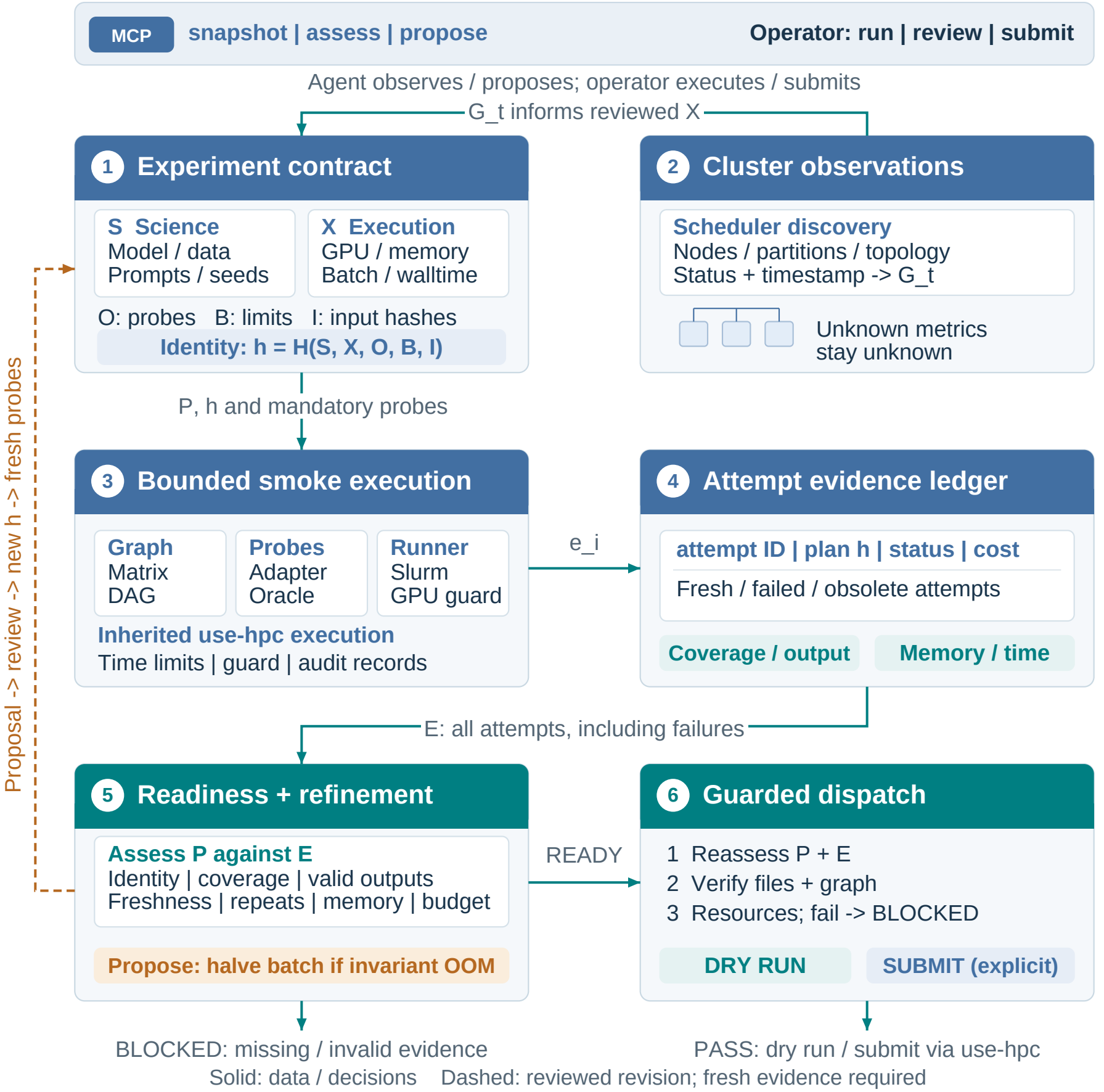
MCP
snapshot | assess | propose
Operator: run | review | submit
Agent observes / proposes; operator executes / submits
G_t informs reviewed X
1 Experiment contract
S Science
Model / data
Prompts / seeds
X Execution
GPU / memory
Batch / walltime
O: probes B: limits I: input hashes
Identity: h = H(S, X, O, B, I)
2 Cluster observations
Scheduler discovery
Nodes / partitions / topology
Status + timestamp -> G_t
Unknown metrics
stay unknown
P, h and mandatory probes
3 Bounded smoke execution
Graph
Matrix
DAG
Probes
Adapter
Oracle
Runner
Slurm
GPU guard
Inherited use-hpc execution
Time limits | guard | audit records
e_i
4 Attempt evidence ledger
attempt ID | plan h | status | cost
Fresh / failed / obsolete attempts
Coverage / output
Memory / time
E: all attempts, including failures
5 Readiness + refinement
Assess P against E
Identity | coverage | valid outputs
Freshness | repeats | memory | budget
Propose: halve batch if invariant OOM
READY
6 Guarded dispatch
1 Reassess P + E
2 Verify files + graph
3 Resources; fail -> BLOCKED
DRY RUN
SUBMIT (explicit)
Proposal -> review -> new h -> fresh probes
BLOCKED: missing / invalid evidence
PASS: dry run / submit via use-hpc
Solid: data / decisions Dashed: reviewed revision; fresh evidence required